

REDIS COMPARIOSN RESULTS

Testing performance optimization with **Redis** by hitting an **API route continuously for 60 seconds** with the “**autocannon**” npm package.

Results: Redis caching **cut latency by more than half, doubled throughput**, and made response times dramatically more predictable. The numbers aren't marginal

This is the difference between an API that struggles under load and one that scales.

WITHOUT REDIS V/S WITH REDIS

Basic Statistics:

=== Basic Statistics ===		=== Basic Statistics ===	
Metric	Count	Metric	Count
-----	-----	-----	-----
Total Requests	26532	Total Requests	61064
Errors	0	Errors	0
Timeouts	0	Timeouts	0
Duration (s)	60.11	Duration (s)	60.13

Without Redis, the system handled **26,532** total requests over **60.11** seconds with zero errors and zero timeouts.

With Redis caching enabled, the same duration (**60.13 seconds**) yielded **61,064** total requests, **a 130% increase in throughput**, while maintaining zero errors and zero timeouts.

Key Takeaway: Handles **130%** more requests

Latency (ms):

=== Latency (ms) ===		=== Latency (ms) ===	
Metric	Value	Metric	Value
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Average	225.46	Average	97.80
Min	102.00	Min	41.00
Max	1002.00	Max	273.00
Std Dev	50.43	Std Dev	29.49

Without Redis, the average latency was **225.46ms**, with a minimum of **102ms**, a maximum of **1002ms**, and a standard deviation of **50.43ms**.

With Redis, average latency dropped to **97.80ms (57% faster)**, minimum fell to **41ms**, maximum to **273ms**, and standard deviation tightened to **29.49ms**.

The Redis implementation is not just faster, it's more consistent.

Key Takeaway: Response is **57%** faster.

Latency Percentiles:

=== Latency Percentiles ===	
Percentile	Value

p0.001	102.00ms
p1	111.00ms
p2.5	132.00ms
p10	192.00ms
p25	201.00ms
p50 (median)	213.00ms
p75	241.00ms
p90	283.00ms
p97.5	339.00ms
p99	390.00ms
p99.9	645.00ms

=== Latency Percentiles ===	
Percentile	Value

p0.001	41.00ms
p1	45.00ms
p2.5	47.00ms
p10	54.00ms
p25	78.00ms
p50 (median)	104.00ms
p75	112.00ms
p90	121.00ms
p97.5	164.00ms
p99	193.00ms
p99.9	260.00ms

At the 99th percentile, latency is 390ms without Redis and 193ms with it, roughly 2x faster for your worst-case users.

Key Takeaway: 2x faster for your **worst-case** users.

Requests/s and Throughput (KB/s):

=== Requests per Second ===	
Metric	Value

Average	442.20
Min	196.00
Max	729.00
Total	26532.00
Sent	26632.00

=== Throughput (KB/s) ===	
Metric	Value

Average	3240.83
Min	1436.50
Max	5342.92
Total (KB)	194455.72

=== Requests per Second ===	
Metric	Value

Average	1017.74
Min	614.00
Max	1900.00
Total	61064.00
Sent	61164.00

=== Throughput (KB/s) ===	
Metric	Value

Average	7458.90
Min	4500.07
Max	13925.29
Total (KB)	447544.26

2.3x More requests and a 2.3x improvement in throughput capacity.

Key Takeaway: 2.3x more data throughput.